Dhruv Prajapati

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Fourth-year Mechanical Engineering student specializing in computational modeling, thermal-fluid systems, and multiphysics simulation, seeking a **Spring 2027/Summer 2027** internship.

Education

Rochester Institute of Technology **Aug. 2023 – Dec. 2027**
Bachelor of Science in Mechanical Engineering, Senior Year, Dean's List 2026, GPA 3.50 Rochester, NY

Technical Skills

Thermal & Fluid Systems: Heat Transfer, Fluid Mechanics, CFD (SolidWorks Flow Simulation, ANSYS Fluent)
CAD Design & Analysis: SolidWorks, ANSYS Mechanical & Fluent, CATIA v5
Programming: Python, MATLAB, Simulink, LabVIEW, Arduino
Hardware & Testing: Hitachi SEM, Oscilloscopes, Keysight Waveform generator, PIV, Instron
Manufacturing: CNC Milling, Lathe, GD&T, 3D Printing, Surface Metrology

Experience

Battery Cell Engineer Intern **Sept. 2026 – Dec. 2026**
TESLA Austin, TX

- **Tabless 4680 Winder** | *SolidWorks, Instrumentation, DOE*
- Analyze encoder and sensor data from 4680 tabless winding equipment to characterize electrode web tension and alignment during jelly-roll formation.
- Evaluate process signals including flag detection, tuck-edge position control, and spool-diameter measurements to understand machine behavior and identify sources of winding variation.

Failure Analysis Engineer Intern **Jun. 2026 – Aug. 2026**
PG&E, Pacific Gas & Electric Company San Ramon, CA

- **Aerodynamics & Structural Multiphysics** | *ANSYS Fluent + Mechanical, MATLAB*
- Developed a multiphysics simulation framework combining transient CFD, structural FEA, and analytical electromagnetic loading to evaluate a 230-kV porcelain post insulator under wind, gravity, and AC-current-induced forces.
- Characterized vortex shedding across 27–54 m/s crossflow using FFT and Strouhal scaling; transferred CFD pressure fields through one-way FSI, predicting > 23 factor of safety.
- **Materials Characterization** | *SEM, Lidar Point Cloud Extraction*
- Investigated a shattered porcelain insulator using SEM-EDS and cantilever testing, identified gunshot-residue metals and ruled out mechanical overloading, confirming ballistic impact as the failure mode.

Mechanical Engineer Researcher **Feb. 2026 – May 2026**
FAST LAB, Fluids Acoustics Structures Laboratory Rochester, NY

- **Test Platform Design** | *Arduino, Pololu Jrk Controller, 3D Printing*
- Designed and built an experimental vortex-ring test platform integrating a linear actuator, Arduino-based control system, and Pololu motor controller to investigate confined-flow fluid dynamics.
- **Vortex-Ring Fluid Dynamics & Heat Transfer** | *Particle Image Velocimetry (PIV)*
- Conducted PIV experiments to quantify vortex-induced flow structures and impingement heat-transfer behavior in confined geometries.

Mechanical Test Engineer Intern **Aug. 2025 – Dec. 2025**
MKS Instruments Rochester, NY

- **Liquid-cooled Heat Sink Analysis** | *SolidWorks Flow Simulation, Experimental test bench*
- Designed and executed experimental characterization of a liquid-cooled microchannel heat sink for a 5 kW pulsed-DC generator (RPG-50), quantifying pressure-drop and flow-rate tradeoffs.
- Validated experimental results using first-principles fluid models and CFD, identifying flow maldistribution and enabling a 25% reduction in hydraulic pressure drop.
- **Cross-functional projects** | *ALT, Engineering Analysis, LabVIEW*
- Supported instrumentation and validation activities across multiple reliability test environments, including HVAC controls, liquid-nitrogen systems, and ALT/HASS/ESS chambers used for hardware qualification.